When Reconstruction Passes:
Natural-Language Autoencoders on VLA Activations Fail Grounding and Semantic Steering

Anonymous Author(s) | Anonymous Institution | anonymous@domain.com

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Abstract. Natural-language autoencoders (NLAs) map neural activations to text and back, offering readable interfaces to model internals. We port the NLA recipe to the GR00T vision-language-action (VLA) backbone: extract per-token hidden states, train an activation verbalizer (AV) and reconstructor (AR), and inject AR outputs as live backbone steers in LIBERO simulation. Supervision uses a multimodal *teacher* (frames + instruction), not human descriptions of what $\mathbf{h}$ encodes—a confound inherited from the original LLM setting. We release `nla-groot` and a three-axis evaluation protocol: (1) offline reconstruction and retrieval, (2) vision-grounded caption judging, (3) closed-loop steer A/B (matched vs. mismatched language). On our main checkpoint, reconstruction metrics pass while grounding, anti-template specificity, and semantic steering fail; aggregate scores **hide** collapse on `image_patch` tokens, where retrieval margin is near zero. **Claim:** NL autoencoders on VLA activations are misleading without vision-grounded and behavioral checks *stratified by token role*—use this protocol before trusting captions or steering policies with language.

Keywords: vision-language-action models, interpretability, activation steering, robot learning

Contributions.

- **Port & tooling:** NLA (AV/AR) on GR00T layer-16 activations with live LIBERO steering hooks and open `nla-groot` pipeline.
- **Three-axis scorecard:** pre-registered reconstruction/retrieval, vision-grounded caption judge, and closed-loop steer A/B (Δ_{cw}).
- **Negative result:** pooled metrics pass while `image_patch` retrieval margin is near chance (0.003), anti-template judge fails (8.3%), and $\Delta_{cw} = 0$—aggregate PASS hides vision-slot collapse.

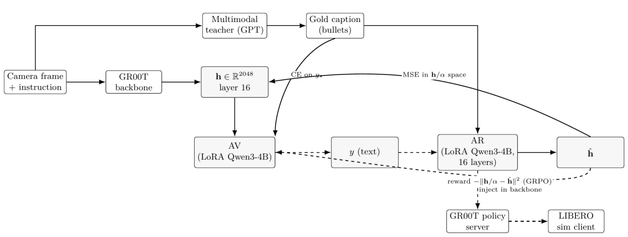

Figure 1. Training (solid) and inference steering (dashed).

Table 1. Headline V3 scorecard (`libero_4suite_v3`).

Metric	Value	Thr.	
retrieval_margin	0.124	0.05	PASS
judge_anti_template	0.083	0.50	FAIL
sim_correct_minus_wrong	0.000	0.05	WARN
`image_patch` margin	0.003	—	near chance

Takeaway. Reconstruction certifies a text–vector codec, not pixel grounding or semantic steering. Stratify evaluation by token role (`image_patch` vs. `last_text`) before deploying NL interfaces on VLAs. Code and scorecard scripts: anonymous repository (placeholder URL for review).